Project: Worked examples
Title: Lifting lugs and spreader beams
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1. Scope and method

Calculates the capacity and utilisation of a pin-loaded lifting lug and of a spreader beam by the BTH-1 approach, the EN 13155 proof route or both. It is a design calculation, not a thorough examination under LOLER or PUWER.

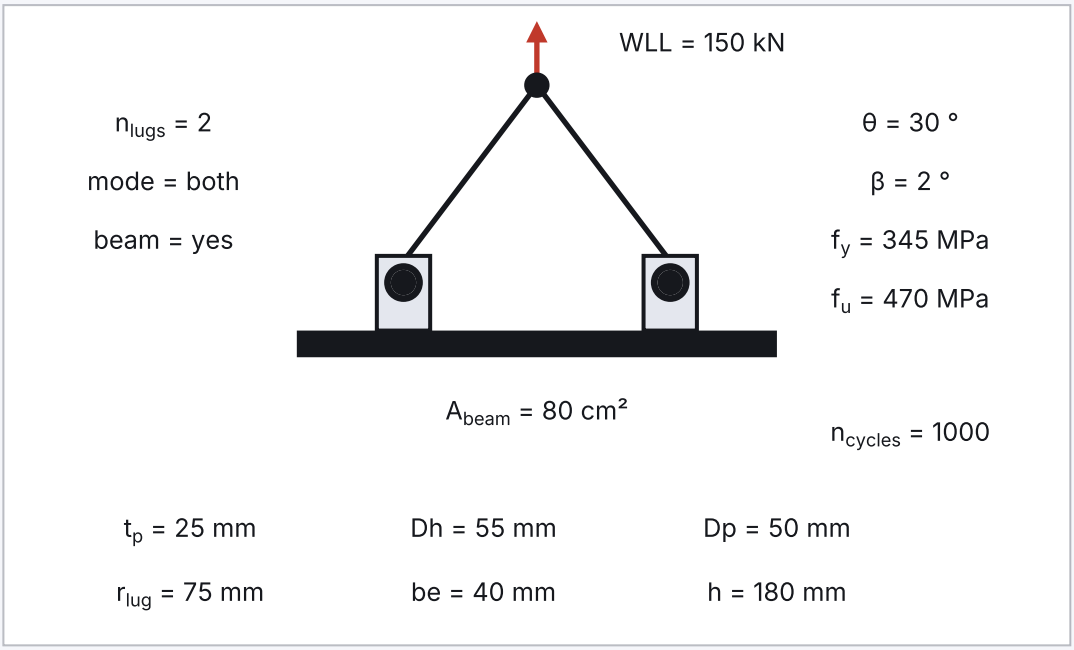


Figure 1 — A hook lifting the rated load WLL through 2 lugs on a spreader beam, each sling at θ from the vertical; each lug a plate t_p with a hole Dh and a pin Dp. Not to scale; the labels are the inputs as entered.

WHAT IT CALCULATES	RELATION
Sling force	$F_v = WLL/n_{eff}$ ($n_{eff} = n_{lugs}$, at most 3); $F_s = F_v/(\cos\theta\cdot\cos\beta)$; $F_{ip} = F_s\cdot\cos\beta$; $F_{op} = F_s\cdot\sin\beta$; $N_{beam} = F_v\cdot\tan\theta$; $M_{op} = F_{op}\cdot h$
Horizontal beam, automatic 6° tilt	pitch ($\theta \pm 6°$, load redistributed, the sling at $\theta - 6°$ takes $2\cdot F_v\cdot\sin(\theta + 6°)/\sin 2\theta$, and the force along the tilted beam is $F_v\cdot\sin(\theta + 6°)/\cos\theta$); roll ($\beta + 6°$); the larger F_s governs, F_{ip}, F_{op} and N_{beam} are the envelopes
Lug section geometry	side = $\min(b_e, r_{lug} - D_h/2)$; $b_{e,eff} = \min(\text{side}, 4t_p, 0.6\cdot(f_u/f_y)\cdot\sqrt{(D_h/\text{side})}\cdot\text{side})$; a_{eff} = a (default $r_{lug} - D_h/2$); $A_t = 2\cdot t_p\cdot b_{e,eff}$; $A_{fr} = [1.13\cdot a_{eff} + 0.92\cdot\text{side}/(1 + \text{side}/D_h)]\cdot t_p$; $\varphi = 55\cdot D_p/D_h$ (degrees); $A_v = 2\cdot t_p\cdot[a_{eff} + (D_p/2)(1 - \cos\varphi)]$; $A_b = D_p\cdot t_p$
Pin clearance	$C_r = 1 - 0.275\cdot\sqrt{(1 - D_p^2/D_h^2)}$
BTH-1 (3-3.3)	$F_a = \min(f_y/n_y, f_u/n_u)$; design load $D = N_d\cdot k_{svc}\cdot F_s$; $P_t = C_r\cdot F_a\cdot A_t$; $P_{sp} = C_r\cdot F_a\cdot A_{fr}$; $P_v = F_a\cdot A_v/\sqrt{3}$; $P_b = k_b\cdot F_a\cdot A_b$; u = D/P for each; dishing $D_h/t_p \le \text{dish}_{max}$

WHAT IT CALCULATES	RELATION
EN 13155 (2× elastic, 3× no release)	P_{EN} = min(Y/2, U/3) for each section, Y with f_y and U with f_u (shear with f/√3; fracture area without C_r); u = F_s/P_{EN}
Lug capacity and proof load	WLL_{cap} = WLL/u_{lug} for the governing section of each mode (smaller of the modes used); beam: u_{beam} from N_{beam} on A_{beam}; proof load = k_{proof}·WLL

Basis: Pin-connected plates (design of lifting lugs)[1](cl. 3-3.3); Non-fixed load lifting attachments: 2× elastic, 3× no release (clause to be confirmed)[2]; Proof of static strength: the calculation route cited, not implemented[3](cl. 5); LOLER 1998: marking of lifting equipment; thorough examination and inspection[4](reg. 7 and 9, 2nd ed.); PUWER 1998: suitability and maintenance of work equipment[5](reg. 4 and 5, 4th ed.).

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
mode	Design mode	both		assumption: Both modes are run so the sheet shows which governs; the owner's code route decides which counts.	any
n_{lugs}	Number of lugs sharing the load (a fourth is not relied on: three share)	2		drawing: LB-150-01 rev A	any
beam	The lugs are on a horizontal spreader or lifting beam (adds the automatic 6° pitch and roll tilt cases)	yes		drawing: LB-150-01 rev A	any
t_p	Lug plate thickness	25	mm	drawing: LB-150-02 rev A	8 … 100 mm
Dh	Hole diameter	55	mm	drawing: LB-150-02 rev A	16 … 300 mm
Dp	Pin diameter (not larger than the hole)	50	mm	drawing: LB-150-02 rev A	12 … 300 mm
r_{lug}	Lug end radius about the hole centre (the plate is twice that wide at the hole)	75	mm	drawing: LB-150-02 rev A	20 … 300 mm
be	Edge distance from the hole edge to the plate edge on each side, across the load line (limited to r_{lug} − $D_h/2$; the effective width for tension is reduced further by $4t_p$ and by f_u/f_y)	40	mm	drawing: LB-150-02 rev A	10 … 200 mm

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
h	Distance from the hole centre to the weld or base, for the out-of-plane moment (optional; blank = no moment given)	180	mm	drawing: LB-150-02 rev A	10 … 1500 mm
A_{beam}	Gross cross-section area of the spreader or lifting beam, for its axial utilisation (optional; blank = lug utilisation only)	80	cm²	drawing: LB-150-03 rev A	10 … 10000 cm²
f_y	Yield strength of the lug plate for its thickness	345	MPa	spec: S355 plate 25 mm, mill certificate MC-2291 (16 mm < t ≤ 40 mm)	235 … 960 MPa
f_u	Ultimate tensile strength of the lug plate	470	MPa	spec: S355 plate 25 mm, mill certificate MC-2291 (lower bound of the ultimate range)	340 … 1100 MPa
WLL	Rated load (working load limit) of the appliance, carried by all its lugs	150	kN	spec: Lifting beam specification LB-150, rated load 150 kN	5 … 5000 kN
θ	Sling angle θ from the vertical, measured in the plane of the lug (the out-of-plane angle β is applied on top: $F_s = F_v/(\cos\theta\cdot\cos\beta)$)	30	°	drawing: LB-150-01 rev A	0 … 45 °
β	Out-of-plane angle β of the sling out of the lug plane	2	°	assumption: Allowance for the sling not hanging exactly in the plane of the lug.	0 … 15 °
n_{cycles}	Load cycles in the life of the lug (the 2×/3× route covers up to 16,000, and beyond that a fatigue assessment is needed and the sheet does not run)	1000		spec: Lifting beam specification LB-150: not more than 1000 lifts in its life	1 … 10000

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
Nd	Design factor of the Design Category (optional; blank = BTH-1 mode is blocked; user record)	1.8		assumption: Illustrative user-record value for this worked example, not a value from any standard: replace with the design factor of your Design Category from your own copy of the code.	1 … 3
n_y	Divisor on the yield strength in the allowable stress (optional; blank = BTH-1 mode is blocked; user record)	1.6		assumption: Illustrative user-record value, not from a standard: replace with your own record.	1 … 3
n_u	Divisor on the ultimate tensile strength in the allowable stress (optional; blank = BTH-1 mode is blocked; user record)	2.2		assumption: Illustrative user-record value, not from a standard: replace with your own record.	1.5 … 5
k_b	Bearing multiple of the allowable stress on the projected pin area (optional; blank = BTH-1 mode is blocked; user record)	1.2		assumption: Illustrative user-record value, not from a standard: replace with your own record.	1 … 2
$dish_{max}$	Dishing-ratio limit on the hole diameter over the thickness (optional; blank = the dishing check is not made; user record)	40		assumption: Illustrative user-record value, not from a standard: replace with your own record.	5 … 60
k_{proof}	Proof-load factor on the rated load (optional; blank = no proof load given; user record)	1.5		assumption: Illustrative user-record value, not from a standard: replace with the proof-load factor of your procedure.	1 … 2

3. Assumptions

A1. This is a design calculation, not a thorough examination. (It checks the lug plate at the pin hole against the user's chosen criteria and nothing else. The owner and the competent person keep the LOLER reg 9 and PUWER duties.)

A2. The sheet screens four sections by nominal stress: tension beyond the hole, single-plane fracture,

shear-out and bearing. (There is no stress-concentration factor, no fatigue (at most 16,000 cycles) and no weld, bolt or beam check. The notch-stress route of EN 13001-3-1 is cited, not implemented.)
A3. The load on the hole is the resultant sling force F_s whatever the out-of-plane angle, which is conservative. (The out-of-plane component and its moment are outputs for the weld and base checks: $F_{op} = F_s \cdot \sin \beta$.)
A4. The lugs share the load equally, and a fourth lug is not relied on. (This is a conservative convention. The share of a statically indeterminate set is not computed, so replace the sharing with the actual share if it is known.)
A5. A horizontal beam adds an automatic 6° tilt as a pitch case and a roll case. (Pitch: the sling angles are θ ± 6° with the load redistributed between the lugs by statics, and the slings meet at the hook. Roll: the out-of-plane angle grows by 6°. N_{beam} is the force along the beam axis, and in the pitch case the more loaded lug sends $F_V \cdot \sin(\theta + 6°)/\cos \theta$ along it. With θ ≤ 6° the pitch case puts the whole load on one lug. With more than two lugs the two-lug redistribution is applied to each lug's share. Owner's specification (6° tilt of a horizontal beam); EN 13155.)
A6. The allowable stress is the one lesser value $\min(f_y/n_y, f_u/n_u)$ for every BTH-1 section. (This follows the published worked example the sheet is checked against. The pin-clearance factor C_r is applied as stated from $D_p/D_h < 1$ with no cut-off. Conservative; owner checklist.)
A7. TempestOS does not authenticate signatures, so any person can switch to another name. (The platform asks for no password, so the author, the two checkers and the competent person are as entered into this installation. ADR-0173; the feature stays off until this is addressed.)
A8. The design factor, service-class factor, divisors, bearing multiple, dishing limit and proof-load factor are the user's records. (They come from the user's own licensed copy of the code. None is shipped and none has a default, and the text and tables of ASME BTH-1 and EN 13155 are not shipped.)
A9. The owner compares the equations in the spec document with BTH-1 and EN 13155 before the feature is enabled. (The single-plane fracture and effective-width relations and the constants 1.13 and 0.92 follow the published course example. The review checklist of the owner test pack lists the comparison.)
A10. mode (Design mode) is taken as both. (Both modes are run so the sheet shows which governs; the owner's code route decides which counts.)
A11. β (Out-of-plane angle β of the sling out of the lug plane) is taken as 2 °. (Allowance for the sling not hanging exactly in the plane of the lug.)
A12. Nd (Design factor of the Design Category (optional; blank = BTH-1 mode is blocked; user record)) is taken as 1.8. (Illustrative user-record value for this worked example, not a value from any standard: replace with the design factor of your Design Category from your own copy of the code.)
A13. n_y (Divisor on the yield strength in the allowable stress (optional; blank = BTH-1 mode is blocked; user record)) is taken as 1.6. (Illustrative user-record value, not from a standard: replace with your own record.)
A14. n_u (Divisor on the ultimate tensile strength in the allowable stress (optional; blank = BTH-1 mode is blocked; user record)) is taken as 2.2. (Illustrative user-record value, not from a standard: replace with your own record.)
A15. k_b (Bearing multiple of the allowable stress on the projected pin area (optional; blank = BTH-1 mode is blocked; user record)) is taken as 1.2. (Illustrative user-record value, not from a standard: replace with your own record.)
A16. $dish_{max}$ (Dishing-ratio limit on the hole diameter over the thickness (optional; blank = the dishing check is not made; user record)) is taken as 40. (Illustrative user-record value, not from a standard: replace with your own record.)
A17. k_{proof} (Proof-load factor on the rated load (optional; blank = no proof load given; user record)) is taken as 1.5. (Illustrative user-record value, not from a standard: replace with the proof-load factor of your procedure.)

4. Working

[19] $n_{eff} = \min(n_{lugs}, 3)$ [6]
$= \min(2, 3)$
$= 2$
Lugs sharing the load (a fourth is not relied on)

[20] $F_v = WLL/n_{eff}$ [6]
$= 150\ kN / 2$
$= 75\ kN$
Vertical load on each lug

[21] $F_{s,nom} = F_v/(\cos\theta \cdot \cos\beta)$ [6]
$= 75\ kN / (\cos 30° \times \cos 2°)$
$= 86.655\ kN$
Sling force on each lug, untilted

[22] $F_{s,pitch} = 2 \cdot F_v \cdot \sin(\theta + 6°)/\sin 2\theta / \cos\beta$ [2]
$= 2 \times 75\ kN \times \sin(36°) / \sin(60°) / \cos(2°)$
$= 101.87\ kN$
Sling force on the more loaded lug when the beam pitches 6°

[23] $F_{s,roll} = F_v/(\cos\theta \cdot \cos(\beta + 6°))$ [2]
$= 75\ kN / (\cos 30° \times \cos 8°)$
$= 87.454\ kN$
Sling force when the beam rolls 6°

[24] $F_s = \max(F_{s,nom}, F_{s,pitch}, F_{s,roll})$ [6]
$=$ governing case: pitch
$= 101.87\ kN$
Sling force on each lug (resultant), governing case

[25] $F_{ip} =$ max over the cases of $F_s \cdot \cos\beta$ [6]
$= 101.87\ kN \times \cos\beta$
$= 101.81\ kN$
In-plane component of the sling force (envelope)

[26] $F_{op} =$ max over the cases of $F_s \cdot \sin\beta$ [6]
$= 101.87\ kN \times \sin\beta$
$= 12.171\ kN$
Out-of-plane component of the sling force (envelope)

[27] $N_{beam} = \max(F_v \cdot \tan\theta, F_v \cdot \sin(\theta + 6°)/\cos\theta)$ [6]
$= 50.904\ kN$, the larger of the level and the pitched case
$= 50.904\ kN$
Axial force along the spreader beam at the more loaded lug (envelope over the cases)

[29] $M_{op} = F_{op} \cdot h$ [6]
$= 12.171\ kN \times 180\ mm$
$= 2.1908\ kN \cdot m$
Out-of-plane moment at the weld or base

[30] side $= \min(b_e, r_{lug} - D_h/2)$ [6]
$= \min(40, 75 - 55/2)\ mm$
$= 40\ mm$
Edge distance on each side of the hole

[31] $a_{eff} = a$, or $r_{lug} - D_h/2$ [6]
$= 75 - 55/2\ mm$
$= 47.5\ mm$
End distance along the load line

[32] $\varphi = 55 \cdot D_p/D_h$ (degrees) [1](3-3.3)
$= 55 \times 50 / 55$
$= 50\ deg$
Angle of the shear planes

[33] $be_{eff} = \min(b_{e,eff}, 4t_p, 0.6 \cdot (f_u/f_y) \cdot \sqrt{(D_h/b_{e,eff})} \cdot b_{e,eff})$ [1](3-3.3)
$= \min(40, 100, 38.339)$ mm
$= 38.339$ mm
Effective width beyond the hole in tension

[34] $A_t = 2 \cdot t_p \cdot b_{e,eff}$ [1](3-3.3)
$= 2 \times 25 \times 38.339$ mm
$= 1917$ mm²
Area beyond the hole in tension

[35] $A_{fr} = (1.13 \cdot a + 0.92 \cdot b_e/(1 + b_e/D_h)) \cdot t_p$ [1](3-3.3)
$= (1.13 \times 47.5 + 0.92 \times 40 / (1 + 40/55)) \times 25$ mm
$= 1874.5$ mm²
Single-plane fracture area beyond the hole (before C_r)

[36] $A_v = 2 \cdot t_p \cdot [a + (D_p/2)(1 - \cos \varphi)]$ [1](3-3.3)
$= 2 \times 25 \times [47.5 + (50/2) \times (1 - \cos 50°)]$ mm
$= 2821.5$ mm²
Shear area of the two shear-out planes

[37] $A_b = D_p \cdot t_p$ [6]
$= 50 \times 25$ mm
$= 1250$ mm²
Projected bearing area of the pin

[38] $Cr = 1 - 0.275 \cdot \sqrt{(1 - D_p^2/D_h^2)}$ [1](3-3.3)
$= 1 - 0.275 \times \sqrt{(1 - (50/55)^2)}$
$= 0.88544$
Pin-clearance factor

[39] $Fa = \min(f_y/n_y, f_u/n_u)$ [6]
$= \min(345 / 1.6, 470 / 2.2)$ MPa
$= 213.64$ MPa
Allowable stress

[40] $D_{bth} = N_d \cdot k_{svc} \cdot F_s$ [1](3-3.3)
$= 1.8 \times 1 \times 101.87$ kN
$= 183.37$ kN
Design load on each lug

[41] $P_t = C_r \cdot F_a \cdot A_t$ [1](3-3.3)
$= 0.88544 \times 213.64$ MPa $\times 1917$ mm²
$= 362.61$ kN
Tension beyond the hole

[42] $P_{sp} = C_r \cdot F_a \cdot A_{fr}$ [1](3-3.3)
$= 0.88544 \times 213.64$ MPa $\times 1874.5$ mm²
$= 354.58$ kN
Single-plane fracture beyond the hole

[43] $P_v = F_a \cdot A_v/\sqrt{3}$ [1](3-3.3)
$= 213.64$ MPa $\times 2821.5$ mm² / $\sqrt{3}$
$= 348.01$ kN
Double-plane shear-out

[44] $P_b = k_b \cdot F_a \cdot A_b$ [1](3-3.3)
$= 1.2 \times 213.64$ MPa $\times 1250$ mm²
$= 320.45$ kN
Pin bearing

[45] $u_t = D/P$ [1](3-3.3)
$= 183.37$ kN / 362.61 kN
$= 0.50568$
Utilisation, tension beyond the hole

[46] $u_{sp} = D/P$ [1](3-3.3)

$= 183.37\ kN / 354.58\ kN$

$= 0.51713$

Utilisation, single-plane fracture

[47] $u_{v} = D/P$ [1](3-3.3)

$= 183.37\ kN / 348.01\ kN$

$= 0.52689$

Utilisation, double-plane shear

[48] $u_{b} = D/P$ [1](3-3.3)

$= 183.37\ kN / 320.45\ kN$

$= 0.5722$

Utilisation, bearing

[49] $u_{bth} = \max(u_{t}, u_{sp}, u_{v}, u_{b})$ [1](3-3.3)

$=$ governing BTH-1 section

$= 0.5722$

BTH-1 utilisation

[50] $cap_{bth} = WLL/u_{bth}$ [1](3-3.3)

$= 150\ kN / 0.5722$

$= 262.14\ kN$

Rated load the lug carries by the BTH-1 approach

[52] $dish = D_{h}/t_{p}$ [6]

$= 55 / 25$

$= 2.2$

Dishing ratio

[53] $u_{dish} = (D_{h}/t)/dish_{max}$ [6]

$= 2.2 / 40$

$= 0.055$

Dishing-ratio utilisation

[54] $P_{en,t} = \min(Y/2, U/3)$ [2]

$= \min(661.35\ kN / 2, 900.97\ kN / 3)$

$= 300.32\ kN$

EN 13155 capacity, tension beyond the hole

[55] $u_{en,t} = F_{s}/P_{EN}$ [2]

$= 101.87\ kN / 300.32\ kN$

$= 0.3392$

EN 13155 utilisation, tension beyond the hole

[56] $P_{en,sp} = \min(Y/2, U/3)$ [2]

$= \min(646.7\ kN / 2, 881.02\ kN / 3)$

$= 293.67\ kN$

EN 13155 capacity, single-plane fracture

[57] $u_{en,sp} = F_{s}/P_{EN}$ [2]

$= 101.87\ kN / 293.67\ kN$

$= 0.34688$

EN 13155 utilisation, single-plane fracture

[58] $P_{en,v} = \min(Y/2, U/3)$ [2]

$= \min(562.01\ kN / 2, 765.63\ kN / 3)$

$= 255.21\ kN$

EN 13155 capacity, double-plane shear

[59] $u_{en,v} = F_{s}/P_{EN}$ [2]

$= 101.87\ kN / 255.21\ kN$

$= 0.39916$

EN 13155 utilisation, double-plane shear

[60] $P_{en,b} = \min(Y/2, U/3)$ [2]
$= \min(431.25\ kN / 2, 587.5\ kN / 3)$
$= 195.83\ kN$
EN 13155 capacity, bearing

[61] $u_{en,b} = F_s/P_{EN}$ [2]
$= 101.87\ kN / 195.83\ kN$
$= 0.52018$
EN 13155 utilisation, bearing

[62] $u_{en} = \max$ over the four sections [2]
= governing EN 13155 section
$= 0.52018$
EN 13155 utilisation

[63] $cap_{en} = WLL/u_{en}$ [2]
$= 150\ kN / 0.52018$
$= 288.36\ kN$
Rated load the lug carries by the EN 13155 route

[64] $lug_{util} = \max$ over the modes used [6]
= governing lug section
$= 0.5722$
Governing lug utilisation

[65] $capacity = WLL/lug_{util}$ [6]
$= 150\ kN / 0.5722$
$= 262.14\ kN$
Rated load the lug carries in the governing mode

[67] $u_{beam} = \max(N_d \cdot k_{svc} \cdot N_{beam}/(A \cdot F_a), \max(2N/(A \cdot f_y), 3N/(A \cdot f_u)))$ [6]
$= N_{beam} = 50.904\ kN$ on $80\ cm^2$
$= 0.053611$
Axial yield utilisation of the beam (buckling: S7)

[68] $util = \max(lug_{util}, u_{beam}, u_{dish})$ [6]
$= \max(0.5722, 0.053611, 0.055)$
$= 0.5722$
Governing utilisation of the lug and the spreader

[70] $proof_{load} = k_{proof} \cdot WLL$ [6]
$= 1.5 \times 150\ kN$
$= 225\ kN$
Proof-load value

[71] $u_{limit} =$ limit
= the utilisation limit
$= 1$
Utilisation limit

5. Results

SYMBOL	VALUE	DESCRIPTION
capacity	262.14 kN	Rated load the lug carries in the governing mode
$governing_{mode}$	"BTH-1: bearing"	Governing mode and section
util	0.57	Governing utilisation of the lug and the spreader
F_s	101.87 kN	Sling force on each lug
F_{ip}	101.81 kN	In-plane component
F_{op}	12.17 kN	Out-of-plane component
N_{beam}	50.90 kN	Horizontal component per lug

SYMBOL	VALUE	DESCRIPTION
cap_{bth}	262.14 kN	Capacity, BTH-1 mode
cap_{en}	288.36 kN	Capacity, EN 13155 mode
M_{op}	2.191 kN·m	Out-of-plane moment
$F_{s,pitch}$	101.87 kN	Sling force, beam pitched 6°
$F_{s,roll}$	87.45 kN	Sling force, beam rolled 6°
side	40 mm	Edge distance on each side of the hole
be_{eff}	38.34 mm	Effective width beyond the hole
a_{eff}	47.50 mm	End distance along the load line
Cr	0.88544	Pin-clearance factor C_r
φ	50°	Shear-plane angle φ
A_v	2821.5 mm²	Shear area
A_t	1917.0 mm²	Tension area beyond the hole
A_{fr}	1874.5 mm²	Fracture area beyond the hole
A_b	1250 mm²	Bearing area
Fa	213.6 MPα	Allowable stress
D_{bth}	183.37 kN	Design load on each lug (governing)
P_t	362.61 kN	Tension beyond the hole
P_{sp}	354.58 kN	Single-plane fracture
P_v	348.01 kN	Double-plane shear
P_b	320.45 kN	Bearing
dish	2.2	Dishing ratio
$P_{en,t}$	300.32 kN	EN 13155 capacity, section t
$P_{en,sp}$	293.67 kN	EN 13155 capacity, section sp
$P_{en,v}$	255.21 kN	EN 13155 capacity, section v
$P_{en,b}$	195.83 kN	EN 13155 capacity, section b
$proof_{load}$	225 kN	Proof-load value

SYMBOL	VALUE	DESCRIPTION
$sheet_{notice}$	"This is a design calculation, not a thorough examination. It does not replace the thorough examination by a competent person (LOLER reg 9) or the marking, testing and certification that the regulations require. Signatures are not authenticated by TempestOS: the signed-in names of the author, the checkers and the competent person are as entered into this installation."	Status of this sheet
checklist	"(1) mark the appliance with its rated load and a unique identification (LOLER reg 7); (2) thorough examination by a competent person before first use and at the intervals required (LOLER reg 9) — this sheet is not one; (3) proof load 225 kN, test certificate where the route requires it; (4) material certificates for the lug plate and the pin, and weld procedure and inspection records for the lug …	Marking and certification checklist
lug_{util}	0.57	Governing lug utilisation
u_{bth}	0.57	Utilisation, BTH-1 mode
u_{en}	0.52	Utilisation, EN 13155 mode
u_{beam}	0.05	Beam axial utilisation
u_{t}	0.51	Utilisation, tension

SYMBOL	VALUE	DESCRIPTION
u_{sp}	0.52	Utilisation, fracture
u_{v}	0.53	Utilisation, shear
u_{b}	0.57	Utilisation, bearing
u_{dish}	0.06	Dishing utilisation
$u_{en,t}$	0.34	EN 13155 utilisation, section t
$u_{en,sp}$	0.35	EN 13155 utilisation, section sp
$u_{en,v}$	0.40	EN 13155 utilisation, section v
$u_{en,b}$	0.52	EN 13155 utilisation, section b

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Design load against the bearing resistance (BTH-1)	183.37 kN	320.45 kN	0.57	[1](3-3.3)
PASS	Design load against the double-plane shear resistance (BTH-1)	183.37 kN	348.01 kN	0.53	[1](3-3.3)
PASS	Sling force against the bearing resistance (3×WLL no release)	101.87 kN	195.83 kN	0.52	[2]
PASS	Design load against the single-plane fracture resistance (BTH-1)	183.37 kN	354.58 kN	0.52	[1](3-3.3)
PASS	Design load against the tension resistance beyond the hole (BTH-1)	183.37 kN	362.61 kN	0.51	[1](3-3.3)
PASS	Sling force against the double-plane shear resistance (3×WLL no release)	101.87 kN	255.21 kN	0.40	[2]

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Sling force against the single-plane fracture resistance (3×WLL no release)	101.87 kN	293.67 kN	0.35	[2]
PASS	Sling force against the tension resistance beyond the hole (3×WLL no release)	101.87 kN	300.32 kN	0.34	[2]
PASS	Dishing ratio against its limit (BTH-1)	2.2	40	0.06	[6]
PASS	Beam axial utilisation against its limit	0.053611	1	0.05	[6]

7. Applicability

WARNING records: tabulated values not taken from a reference record are f_y, f_u (Material properties); Nd (Design factor and service-class factor); n_y, n_u, k_b (Allowable-stress divisors and bearing multiple); $dish_{max}$, k_{proof} (Dishing limit and proof-load factor). Enter each as one so its source, edition and licence travel with it. [record]
WARNING k_{svc}: no service-class factor is entered, so 1.0 is used. Enter it from your own copy of the code if it applies. [k-svc]
WARNING beam: the 6° tilt cases are applied, a pitch of θ ± 6° and a roll of β + 6°. The larger sling force governs. [tilt]
WARNING be: the effective width 38.339 mm is below the edge distance 40 mm, so the tension area uses the smaller. [be-eff]

8. Verification

```
Template struct.lifting@1.2.0 Content hash 249b820c0d253d6b Engine calc-engine/1.0.0
Known-answer cases passed 5/5 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
```

9. References

1. American Society of Mechanical Engineers. Design of Below-the-Hook Lifting Devices. ASME BTH-1. ASME.
2. BS EN 13155.
3. BS EN 13001-3-1.
4. Health and Safety Executive, Safe use of lifting equipment (L113).
5. Health and Safety Executive, Safe use of work equipment (L22).
6. spec.